

Internship Management System

Student Module — Full Functional Overview

Tech Stack: MERN (MongoDB · Express.js · React.js · Node.js)

1. Project Overview

This document provides a comprehensive functional and technical overview of the Student Module within the Internship Management System. The system is a collaborative academic project; the authentication layer (sign-up and login) is handled by a separate team member, while the module described here covers everything the student interacts with after successfully logging in.

Field	Detail
Project Name	Internship Management System
Module Scope	Student-facing functionality post-login
Tech Stack	MongoDB, Express.js, React.js, Node.js (MERN)
Out of Scope	Student sign-up / login, Department approval panel
Document Purpose	Full overview of student module flows and features

2. Module Boundaries

The student module begins immediately after a student authenticates. The following table clarifies what is and is not part of this module.

✓ Included in This Module	✗ Out of Scope
• Internship Status navigation	• Student sign-up and login

• Selection status declaration	• Department approval workflow
• Offer letter submission	• Department notification handling
• Progress tracker (Your Progress)	
• Record Book management	
• Weekly log entry and skill tracking	

3. Navigation Bar Structure

After login, the student sees the following navigation items. Note that Record Book is always visible but remains disabled until the required conditions are met.

Nav Item	Default State	Enabled When
Internship Status	Enabled	Always accessible after login
Record Book	Disabled (visible)	Department approves the internship

4. Internship Status Page

When the student clicks Internship Status in the navigation bar, they are taken to a page presenting three action options. Each option triggers a distinct workflow.

Button / Option	Purpose
I Have Got Selected	Triggers the offer letter submission workflow for students who secured an internship placement.
Not Selected Yet	Records that the student has not yet secured placement; pauses the formal approval process.
Your Progress	Displays a milestone tracker showing the current status of the student's internship journey.

5. Flow A — Student Selected for Internship

This is the primary happy-path workflow for a student who has been accepted by a company. Each step must be completed in sequence. Skipping any step triggers an appropriate error message.

5.1 Step-by-Step Flow

1	Click “I Have Got Selected” The student clicks this button on the Internship Status page. The system unlocks the “Formalize Your Offer” box, which was previously greyed out and unavailable.
2	Formalize Your Offer — Auto-mark Placement Search As soon as the student clicks “I Have Got Selected,” the Placement Search milestone under Your Progress is automatically marked as complete. No manual action is needed.
3	Fill in Offer Details The student completes the Formalize Your Offer form with: Company Name, Job Position / Role Title, Internship Start Date, and Upload of the Offer Letter (PDF or image).
4	Submit the Offer Letter On submission, the system validates all required fields. If any field is missing, an error message is shown and submission is blocked. On success, the “Pending Approval” milestone under Your Progress is automatically marked, and a notification is sent to the Department for review.
5	Department Approval (External) The Department reviews the submitted offer letter and approves or rejects it. This action is performed in the Department module, which is outside this student module’s scope.
6	Approved Milestone Auto-marked Once the Department approves the internship, the Approved milestone in the student’s Your Progress tracker is automatically marked as complete.
7	Go to Record Book Button Enabled After approval, a “Go to Record Book” button becomes active. Clicking it navigates the student to the Record Book page and simultaneously enables the Record Book item in the navigation bar.

5.2 Formalize Your Offer — Form Fields

Field	Type	Validation Notes
Company Name	Text Input	Required; cannot be empty
Job Position	Text Input	Required; cannot be empty
Internship Start Date	Date Picker	Required; must not be in the past
Offer Letter	File Upload	Required; accepted formats: PDF, JPG, PNG

6. Flow B — Student Not Selected Yet

If a student has not yet secured an internship placement, they select the “Not Selected Yet” option on the Internship Status page. This flow is simpler and does not involve offer letter submission.

1

Click “Not Selected Yet”

The student selects this option to inform the system of their current placement status.

2

Placement Search Auto-marked

Just like in Flow A, the Placement Search milestone under Your Progress is automatically marked as complete to record that the student has acknowledged their status.

3

Process Paused Message Displayed

A clear on-screen message is shown to the student: “Process Paused.” This informs them that no further approval steps can proceed until they secure a placement.

4

Department Notification (External)

A “Not Selected” notification is automatically sent to the Department module. The Department can then follow up or provide guidance. Handling on the department side is out of scope.

Important: In Flow B, the Formalize Your Offer box remains disabled. The student cannot proceed to Pending Approval, Approved, or Record Book without first completing Flow A.

7. Your Progress — Milestone Tracker

The Your Progress section visually displays the student's journey through the internship process as a series of milestones. Each milestone is automatically marked by system events; students cannot manually check them.

#	Milestone	Triggered By	Applies To
1	Placement Search	Clicking "I Have Got Selected" OR "Not Selected Yet"	Both Flow A and Flow B
2	Pending Approval	Submitting the offer letter form successfully	Flow A only
3	Approved	Department approves the internship in their module	Flow A only

8. Record Book — Weekly Internship Log

The Record Book is the student's digital internship journal. It is unlocked only after the internship is approved by the Department. Each week during the internship, the student must add a new log entry.

8.1 Access Conditions

- The "Go to Record Book" button must be clicked from the Your Progress section after the Approved milestone is marked.
- The Record Book navigation item becomes enabled upon first access.
- Accessing Record Book directly (without following the approval flow) is blocked and shows an error.

8.2 Weekly Log Entry Fields

Each log entry corresponds to one week of the internship. All fields listed below must be completed before the entry can be saved.

Field / Section	Input Type	Description
Week Number	Number Input	Auto-increments; identifies the week of internship (Week 1, 2, 3...)
Reporting Period	Text / Label	Descriptive label for the reporting period (e.g., "Week of June 10")
Start Date	Date Picker	The start date of the reporting week
End Date	Date Picker	The end date of the reporting week; must be after Start Date
Tasks Completed	Text Area	A detailed description of all tasks the student completed that week
Challenges Encountered	Text Area	Any difficulties, blockers, or unexpected problems faced during the week
Reflection and Learning Outcomes	Text Area	What the student learned; personal or professional growth observations
Skills Gained This Week	Multi-select + Add	Student selects from a predefined skill list; can also add custom skills via an "Add Skill" option

8.3 Skills Gained This Week — Feature Detail

This field supports two interaction modes:

- Select from existing skills: A multi-select dropdown or tag interface populated with a predefined skill library.
- Add a new skill: An "Add Skill" button allows the student to type and save a custom skill not in the existing list. The new skill is then available for selection in future entries.

9. Error Handling and Validation Rules

The system enforces a strict step-by-step flow. If any required step is skipped or any field is left incomplete, a descriptive error message must be displayed to the student. Below are the key scenarios.

Scenario	Expected System Response
Student tries to access Formalize Your Offer without clicking "I Have Got Selected"	Box remains disabled; error message if forced access is attempted

Student submits the offer form with missing required fields	Submission blocked; inline error messages highlight empty fields
Student tries to access Record Book before approval	Navigation disabled; error: "Record Book is not available yet."
Student tries to skip Pending Approval (no offer letter submitted)	Approved milestone cannot be triggered; department never receives notification
Record Book entry saved without filling all required log fields	Save blocked; error messages displayed next to empty fields
End Date is set before Start Date in Record Book	Validation error: "End date must be after start date."

10. End-to-End Flow Summary

The diagram below represents the complete student journey in sequential order.

Flow A — Internship Secured

1. Student logs in → clicks Internship Status
2. Clicks "I Have Got Selected" → Formalize Your Offer box is unlocked
3. Placement Search milestone auto-marked in Your Progress
4. Student fills in Company Name, Job Position, Start Date, and uploads Offer Letter
5. Student submits form → Pending Approval milestone auto-marked
6. Notification sent to Department (external action)
7. Department approves → Approved milestone auto-marked
8. Go to Record Book button becomes active
9. Student clicks Go to Record Book → Record Book page opens; nav item enabled
10. Student fills weekly log entries (Week, Period, Dates, Tasks, Challenges, Reflections, Skills)

Flow B — Not Yet Selected

11. Student logs in → clicks Internship Status
12. Clicks "Not Selected Yet"
13. Placement Search milestone auto-marked in Your Progress
14. Process Paused message displayed to student
15. Department receives "Not Selected" notification (external)
16. Record Book remains disabled; student cannot proceed further until placement is secured

11. MERN Stack Implementation Notes

The following section provides high-level guidance on how this module maps to the MERN stack. These are implementation suggestions for the development team.

11.1 MongoDB — Suggested Collections

- `students` — stores student profile data (linked from auth)
- `internship_applications` — stores offer letter details, company name, position, start date, file reference
- `progress_status` — tracks milestone states per student (placementSearch, pendingApproval, approved)
- `record_book_entries` — stores weekly log entries per student
- `skills` — stores the master list of available skills; new custom skills are added here

11.2 Express.js — Key API Endpoints

Method	Endpoint	Purpose
POST	<code>/api/internship/selected</code>	Record selection status; trigger Placement Search milestone
POST	<code>/api/internship/not-selected</code>	Record not-selected status; trigger Placement Search milestone
POST	<code>/api/internship/offer</code>	Submit offer letter form with file upload
GET	<code>/api/internship/progress/:studentId</code>	Fetch current milestone states for a student
POST	<code>/api/recordbook/entry</code>	Add a new weekly log entry
GET	<code>/api/recordbook/:studentId</code>	Fetch all log entries for a student
GET	<code>/api/skills</code>	Fetch available skills list
POST	<code>/api/skills</code>	Add a new custom skill

11.3 React.js — Key Components

- `InternshipStatusPage` — renders the three option buttons
- `FormalizeOfferBox` — conditionally enabled form component with file upload
- `ProgressTracker` — milestone display component; reads milestone state from backend

- RecordBookPage — weekly log list and entry form
- WeeklyLogForm — controlled form for all log fields including skill selector
- SkillSelector — multi-select component with Add Skill capability
- NavBar — conditionally enables/disables the Record Book link based on approval state

11.4 Node.js / File Handling

- Multer middleware handles offer letter file uploads.
- Files stored in a /uploads directory or a cloud storage service (e.g., AWS S3).
- File references (paths or URLs) saved in the internship_applications collection.

12. Glossary

Term	Definition
Internship Status	The primary navigation section students use to declare their placement situation.
Formalize Your Offer	A form box that becomes active once a student declares they have been selected; used to submit placement details.
Your Progress	A milestone tracker showing the stages of the student's internship approval journey.
Placement Search	The first milestone, auto-marked when the student declares their selection status.
Pending Approval	The second milestone, auto-marked when the offer letter is submitted.
Approved	The third milestone, auto-marked when the Department approves the internship.
Record Book	The student's weekly internship journal, enabled only after approval.
Go to Record Book	A button that appears after the Approved milestone and navigates to the Record Book page.
Process Paused	A system message shown when a student declares they are not yet selected.
Skills Gained This Week	A multi-select + add field in each weekly log entry for recording skills acquired.
MERN	MongoDB, Express.js, React.js, Node.js — the full-stack JavaScript framework used for this project.

